Week 1:
Introduction to Machine Learning

WIA1006 | Semester 2 | Tutorial Session | University of Malaya

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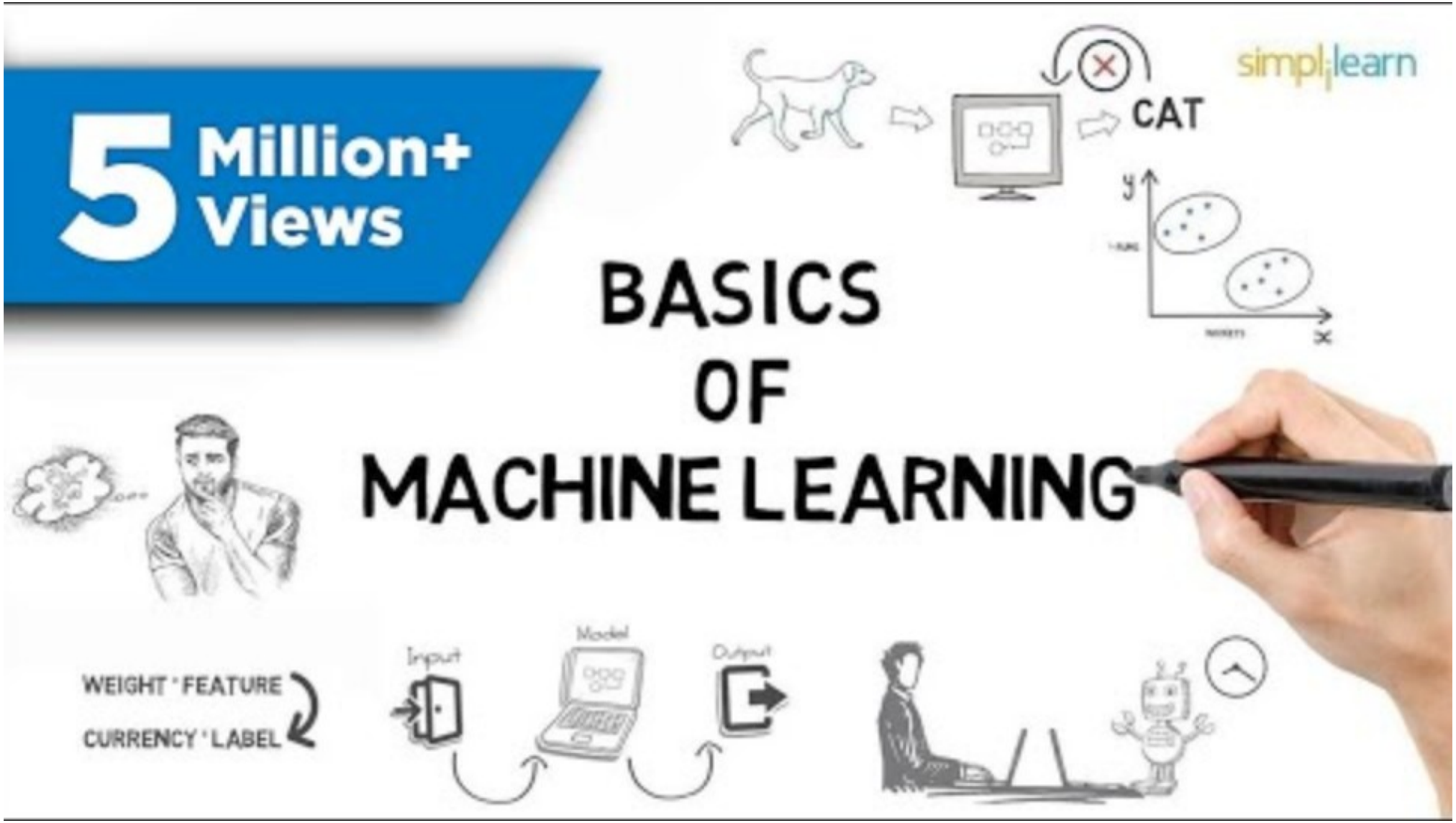
What is Machine Learning?

HOW AI WORKS
MACHINE LEARNING
CODE

What is Machine Learning?

- *"Machine learning lets computers learn patterns from data and improve predictions or decisions without being explicitly programmed for every case."*

- *Examples:*
  - *Traffic prediction*
  - *Image recognition*
  - *Recommendation systems*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BASICS OF MACHINE LEARNING
WEIGHT · FEATURE
CURRENCY · LABEL
Input
Model
Output

Basics of Machine Learning?

- **Programming Vs Machine Learning**:
  - ***Traditional Programming***: *Rules + Data -> Output*
  - ***Machine Learning***: *Data + Output -> Rules (Patterns)*

- **Types of Machine Learning:**
  - ***Supervised Learning***: *Learns from labelled examples*
  - ***Unsupervised Learning***: *Finds hidden patterns*
  - ***Reinforcement Learning***: *Learns by trial and error*

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Question 1

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v1 (latest)

A computer program is said to learn from experience, E, with respect to some task, T, and some performance measure, P. If its performance on T, as measured by P, improves with experience E. Suppose we feed a learning algorithm a lot of historical trajectory data of vehicles, and have it learn to predict traffic speed. In this setting what is E?

Answer:

Next page

Discussions?

Question: Identify T, P and E in this machine learning example. Predict traffic speed?

T Predict traffic speed	**P** Prediction accuracy	**E** Historical data

The model studies many past vehicle movement records. These past records are its **experience**. By **learning** from them, it gets better at **predicting future** traffic speed.

Answer: E is the process of the algorithm examining a large amount of historical trajectory data.

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Question 2

Not yet answered

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Suppose you are working on traffic prediction, and you would like to predict whether traffic will be heavy at 5pm tomorrow or not. You want to use a learning algorithm for this. Would you treat this as a classification or a regression problem?

Answer:

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Discussions?

Question: Will traffic be heavy at 5 pm tomorrow?

Is this classification or regression?

Classification

Predicts a label or category

Possible outputs:
- Heavy
- Not Heavy

✓

This problem fits here

Regression

Predicts a continuous number

Example output:
Traffic speed = 18.5 km/h

Not this one

Answer: This is a **classification** problem because the output is a label: **Heavy / Not Heavy**

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Question 3

Not yet answered

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v1 (latest)

Take a collection of 1000 essays written on the US Economy, and find a way to automatically group these essays into a small number of groups of essays that are somehow "similar" or "related".

- a. Supervised
- b. Unsupervised

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Discussions?

Question: What kind of learning is this?

- 1000 essays on the US Economy
- Automatically group similar essays
- Type of learning: **Unsupervised**

The model is not given any correct **labels** in advance. It simply looks for **patterns** and groups essays that seem similar.

Tip: Grouping similar data without known labels = Unsupervised learning

Answer: b. Unsupervised

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Question 4

Not yet answered

Flag question

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v1 (latest)

Given 50 articles written by male authors, and 50 articles written by female authors, learn to predict the gender of a new manuscript's author (when the identity of this author is unknown).

- a. Supervised
- b. Unsupervised

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Discussions?

Question: Is this supervised or unsupervised?

- Training data has known labels
- 50 articles labeled male
- 50 articles labeled female
- Type of learning: **Supervised**

Because the model learns from examples that already come with the correct labels, this is a supervised learning problem.

Tip: Known labels in the training set = Supervised learning

Answer: a. Supervised

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Question 5

Not yet answered

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v1 (latest)

Examine a large collection of emails that are known to be spam email, to discover if there are sub-types of spam mail.

- a. Supervised
- b. Unsupervised

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Discussions?

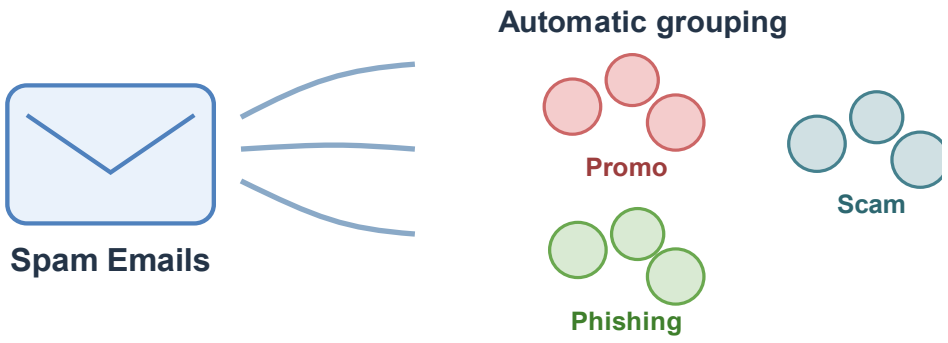
Question: Examine a large collection of emails that are known to be spam email, to discover if there are sub-types of spam mail.

Answer: b. Unsupervised

Why is it unsupervised?

- All emails are already known to be spam.
- The goal is not to predict “spam” vs “not spam”.
- The goal is to discover hidden groups inside spam emails.
- No subgroup labels are given in advance.

Visual intuition



Simple idea: Find natural sub-groups within spam emails without giving the model the exact sub-type labels.

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Question 6

Not yet answered

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v1 (latest)

Examine a web page, and classify whether the content on the web page should be considered "child friendly" (e.g., non-pornographic, etc.) or "adult."

- a. Supervised
- b. Unsupervised

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Finish attempt ...

Discussions?

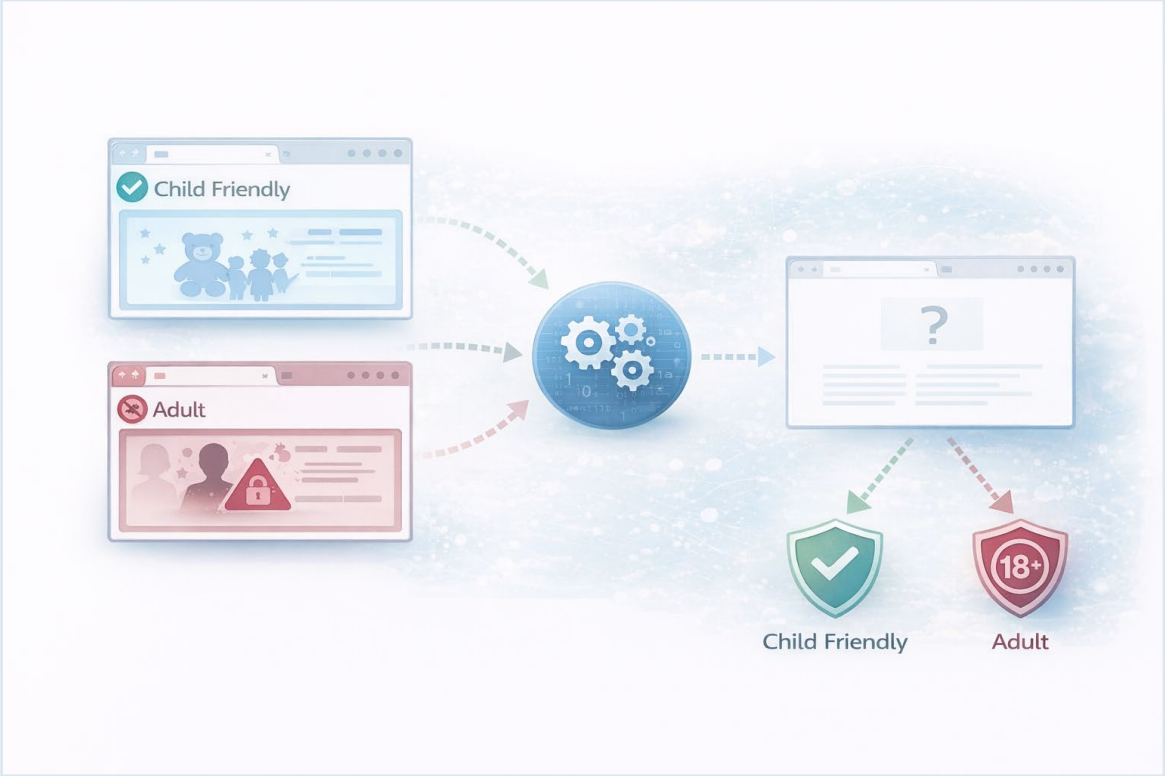
Should a web page be labeled child friendly or adult?

- This is a classification task
- Possible labels: child friendly or adult
- The model learns from pages with known labels
- Type of learning: **Supervised**

Because the training examples already have the correct answers, the model learns to predict the label for a new web page.

Tip: Known labels in training data = Supervised learning

Answer: a. Supervised



Q & A

Thank You!

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